CSE4203: Principles of Software Architecture

# FanZone

Football Match Ticketing System for the Egyptian League

## Milestone 1: Proposal & Architectural Requirements

| Field | Value |
|---|---|
| Project name | FanZone |
| Team name | AAZIYR |
| Milestone | M1: Proposal & Architectural Requirements |
| Course | CSE4203: Principles of Software Architecture |
| Submission date | October 23rd, 2026 |

### Team Members

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# Contribution Table

Each member's contribution to this milestone.

| Member | Sections / Tasks |
|---|---|
| Ziad Khaled | Section 1 System Overview; Section 2 Stakeholders and Concerns |
| Israa Amr | Section 3 System Context (C4 L1 diagram) |
| Youssef Mohamed | Section 4 Functional Requirements |
| Anas Mostafa | Section 5 Quality Attribute Scenarios |
| Retaj Amr | Section 6 Constraints |
| Abdullah Khaled | Section 7 ASRs and Utility Tree; formatting |

**Project proposal:**

FanZone is a web platform for buying Egyptian league football tickets. Today, fans face unannounced ticket releases, websites that crash under heavy traffic, no way to buy tickets for friends, no reward for loyal attendees, and paper tickets they must print. In FanZone, verified fans browse matches and stadiums, book tickets by category, pay online, and receive an e-ticket by email that is scanned at the stadium gate. A fan may buy one ticket for themselves and can buy for friends only when each friend approves the request. The proposal meets the project requirements as follows:

1. **User roles.** The system has five distinct user roles:
    - **Fan:** browses, books, pays, and requests or grants purchase authorizations.
    - **Admin:** manages matches, stadiums, release schedules, refunds and logs.
    - **Customer Support:** resolves complaints and assists with refunds.
    - **Marketing Team:** runs promotions and loyalty campaigns using consented data only.
    - **Gate Operator:** scans tickets at the stadium gate.
2. **Competing quality attributes.**
    - **Performance vs. consistency:** up to 200,000 fans booking at once during a ticket release must be served quickly, yet a category must never be sold more tickets than it holds.
    - **Security vs. usability:** identity checks and friend approvals protect fans and reduce scalping, but they add steps to a time-critical purchase.
3. **External system integrations.** FanZone integrates with four external systems:
    - a **payment gateway** for payments and refunds
    - an **email service** for e-tickets and authorization requests
    - the **EPFCA official match data source** for schedules, stadiums and prices
    - a **stadium Gate Validation System** for ticket scanning
4. **Separated parts.** The system is split into two separately deployable parts:
    - a **web frontend** for fans and staff
    - a **backend API** for booking, authorization, payment and ticketing
5. **Realistic scope.** The Milestone 3 prototype will implement only the core flows:
    - registration, browsing and booking
    - friend authorization
    - payment through the gateway’s sandbox
    - e-ticket delivery by email
    - a simulated gate scan

# 1. System Overview

## 1.1 Problem

Buying football match tickets in Egypt is frustrating for fans. Ticket releases follow no fixed or announced schedule, so fans must keep repeatedly checking the ticketing website to check whether tickets have been released, and many miss out simply because they checked at the wrong time.

When tickets for high-demand matches are released, the website often crashes under heavy traffic. Fans cannot complete their purchases, and tickets go to whoever happens to get through rather than to the fans who submitted their requests first.

Fans can also only buy tickets for themselves. Friends or family members who want to attend together must each compete for tickets separately during the same crowded release, so groups are often split up or some members miss the match entirely.

There is also no loyalty system to reward committed supporters. A fan who attends every match gets no priority for important matches over someone who attends only once.

Finally, after buying a ticket, fans must print it before going to the stadium, which adds an extra step and is inconvenient for those without easy access to a printer.

## 1.2 Proposed solution

**FanZone** is a web platform where verified fans log in, browse matches and stadium details, book tickets, pay online, and receive a digitally signed e-ticket by email. The ticket is shown on the fan's phone and scanned at the stadium gate by the Gate Validation System, so no printing is needed. Admins publish ticket release schedules in advance, so fans know when tickets go on sale, and the platform is designed to remain stable and responsive under peak booking loads instead of crashing.

Its defining feature is **delegated group purchasing**: a fan may buy only one ticket for themselves and may buy tickets for friends only when each friend explicitly authorizes the purchase. Authorization requests can be approved or rejected; checkout cannot be completed without valid authorization. Each ticket is bound to an identified person, which reduces scalping and makes every ticket traceable.

## 1.3 Users

- **Fans** browse, book, pay, and request or grant purchase authorizations.
- **Admins** manage matches, stadiums, ticket release schedules, refunds and system logs.
- **Customer Support Staff** resolve complaints and assist with refunds and ticket issues.
- **Marketing Team** runs promotions, loyalty campaigns and match predictions using consented data only.
- **Gate Operators** scan tickets at the stadium through the Gate Validation System.

## 1.4 Why FanZone is architecturally interesting

- **Flash-crowd demand vs. strict consistency.** Popular releases cause sudden spikes of up to 200,000 concurrent fans, yet the number of tickets sold in each category must never exceed that category's capacity. Handling such high traffic requires the system to scale efficiently while maintaining accurate ticket availability and preventing overselling.

- **Security and privacy vs. usability.** Identity binding, friend authorization and marketing consent protect fans, but every extra step can slow down or frustrate them during a time-limited release.
- **Distributed, partially offline operation.** Gates must keep validating tickets during network outages while still detecting duplicates and forgeries.
- **Dependence on external systems** (payment gateway, email, EPFCA data, gate system) whose failures must not cause lost payments or inconsistent bookings.
- **Time-bound workflows.** Tickets reserved during checkout must be released back to their category if the fan does not complete payment in time. This requires reliable timers and correct state changes while thousands of fans book at once.

# 2. Stakeholders and Concerns

| ID | Stakeholder | Role / interest | Key concerns | Related QAs |
|---|---|---|---|---|
| SH-01 | Fan / End user | Books tickets, views match and stadium details, buys for friends via authorization, approves friends' requests. | Fair access during rushes; fast booking; clear authorization flow; secure payment; ticket arrives reliably; privacy of personal data. | Performance, usability, security, availability |
| SH-02 | Admin | Manages matches, stadiums, ticket release schedules, refunds and system logs. | Correct capacities and prices; controlled release timing; complete audit logs; easy data correction. | Consistency, auditability, usability |
| SH-03 | Marketing Team | Run promotions, loyalty campaigns and match predictions. | Campaign reach and analytics; must not access personal data without consent; easy campaign changes. | Security (confidentiality) , modifiability |
| SH-04 | Customer Support Staff | Resolves complaints; assists with refunds and ticket issues. | Quick lookup of bookings, payments and authorizations; clear transaction history; role-limited access. | Usability, auditability, security |
| SH-05 | EPFCA (Egyptian Professional Football Clubs Association) | Publishes official match schedules, stadium info and pricing. | Sales follow official data and timelines; national sports regulations respected; schedule changes reflected promptly. | Interoperability, consistency, compliance |
| SH-06 | Maintenance / IT Team | Keeps the system stable, fixes crashes, handles updates. | Monitoring and diagnosability; deployments within limited downtime windows; limited budget and infrastructure. | Availability, deployability, cost |
| SH-07 | Ticket Validator (Gate System and gate operators) | Scans tickets at stadium gates. | Scan speed; offline tolerance; detection of duplicate or fake tickets; simple operator feedback. | Performance, availability, security |
| SH-08 | Payment Gateway / Bank | Processes payments and refunds; verifies transactions (also an external system). | Secure, standards-compliant integration; successful transaction verification; operation within bank/network availability and security rules. | Security, interoperability, availability |
| SH-09 | Developers | The team that builds and improves FanZone. | Clear module boundaries; testable code; replaceable integrations; realistic scope. | Modifiability, testability |
| SH-10 | Business Owner / FanZone Management | Owns the product, revenue and reputation. | No overselling or fraud incidents; low operating cost; growth through marketing; regulatory compliance. | Consistency, security |
| SH-11 | Stadium Security / Venue Management | Controls entry and crowd safety at stadiums. | Attendance never exceeds capacity; each attendee identifiable; entry not blocked by system failure. | Consistency, availability |
| SH-12 | Data-protection regulator | Oversees handling of personal data under applicable Egyptian law. | Lawful consent; data minimization; protection of identity data; breach handling. | Security (confidentiality), compliance (C-08) |

# 3. System Context (C4 Level 1)

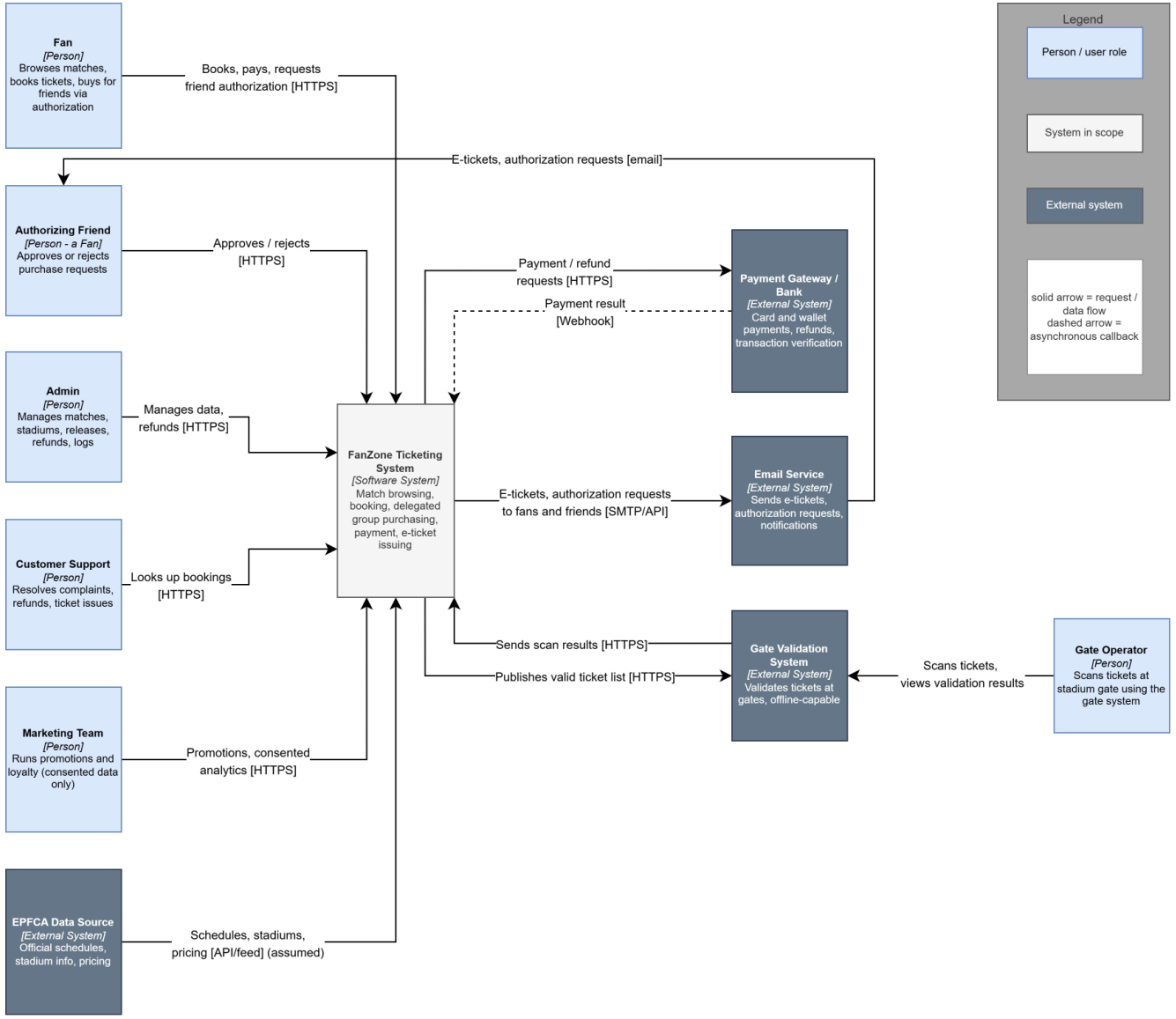



*Figure 1. FanZone system context diagram (C4 Level 1) [2]: FanZone as a single system, its users and external systems. Source: docs/m1/diagrams/fanzone_context.mmd.*

# 4. Functional Requirements

Requirements are written as user stories and prioritized with MoSCoW: **M** = Must, **S** = Should, **C** = Could. The **M3** column marks what the Milestone 3 prototype will implement: **Yes or No.**

## 4.1 User stories

| ID | Actor | User story | MoSCoW | M3 |
|---|---|---|---|---|
| FR-01 | Fan | As a fan, I want to register with my national ID/passport, mobile number and email, and set my marketing consent, so that I can buy tickets bound to my identity. | M | Yes |
| FR-02 | All users | As a user, I want to log in and out securely, with access limited to my role (Fan, Admin, Support, Marketing). | M | Yes |
| FR-03 | Fan | As a fan, I want to browse upcoming matches and view stadium details, ticket categories, prices and availability. | M | Yes |
| FR-04 | Fan | As a fan, I want to select a ticket category and have my tickets held while I complete checkout (10-minute hold, assumed). | M | Yes |
| FR-05 | Fan | As a fan, I want to buy one ticket for myself and be blocked from buying a second one without friend authorization. | M | Yes |
| FR-06 | Fan | As a fan, I want to send an authorization request to a registered friend so I can buy a ticket on their behalf. | M | Yes |
| FR-07 | Friend (Fan) | As a friend, I want to approve or reject an authorization request from a link sent to my email or my account. | M | Yes |
| FR-08 | System rule | As the business owner, I want checkout to re-validate all authorizations, so no ticket is issued without consent. | M | Yes |
| FR-09 | Fan | As a fan, I want to pay online through the payment gateway and see a clear success or failure result. | M | Yes |
| FR-10 | Fan | As a fan, I want to receive each ticket by email, so I don’t need to print it. | M | Yes |
| FR-11 | Gate Operator | As a gate operator, I want each scanned ticket validated (valid, already used, invalid) quickly, including when the gate is offline. | M | Yes |
| FR-12 | Admin | As an admin, I want to manage matches, stadiums, ticket categories, capacities and release schedules, and view system/audit logs. | M | Yes |
| FR-13 | Admin / EPFCA | As an admin, I want official schedules, stadium data and prices imported from the EPFCA data source, so FanZone matches official information. | S | Yes |
| FR-14 | Support / Admin | As support staff, I want to look up a fan’s bookings, payments and authorizations, handle complaints, and start refunds approved by an admin. | S | No |
| FR-15 | Marketing | As the marketing team, I want to run promotions, loyalty campaigns and match predictions targeting only fans who consented. | C | No |

# 5. Quality Attribute Scenarios

### QAS-01: Ticket-release rush

| Part | Performance |
|---|---|
| Source | External: fans using the FanZone website. |
| Stimulus | Demand Spikes: 200,000 concurrent fans submitting booking requests within the first 10 minutes of ticket sales opening for a high-demand match. |
| Artifact | System services: booking and ticket-inventory services. |
| Environment | Overload: peak load during a high-demand match ticket release. |
| Response | Booking requests are processed, tickets are reserved for successful requests, and each fan receives a booking confirmation or a clear “sold out” notification. When a category reaches full capacity, all subsequent requests for that category are immediately rejected. |
| Response measure | Latency: 95% of booking requests receive a response in under 2s and 99% in under 5s with 200,000 concurrent users.<br>Error rate: fewer than 0.5% of requests fail with a server error.<br>Data loss: zero accepted bookings lost. |

### QAS-02: Payment gateway outage

| Part | Availability |
|---|---|
| Source | External system: Payment Gateway / Bank. |
| Stimulus | Fault (omission): the gateway stops responding or times out during checkout, for up to 15 minutes. |
| Artifact | Communication channel between FanZone and the payment gateway; checkout process. |
| Environment | Normal operation or overloaded (during a release). |
| Response | Detect the fault, log it, and degrade gracefully: Stop sending new payment requests. Extend the holds of affected fans until the gateway recovers (up to 15 minutes beyond the normal 10-minute hold) and tell them to retry. Recover uncertain payments by checking their status with the gateway once it is back. |
| Response measure | Detection time: within 30s.<br>Time in degraded mode: ≥ 2minutes.<br>Service during degraded mode: browsing and ticket holds stay available (≥ 99% of requests succeed).<br>Faults handled: 100% of uncertain payments resolved within 15 minutes of gateway recovery, with zero charged-but-unbooked or booked-but-unpaid orders. |

### QAS-03: Gate network loss

| Part | Availability |
|---|---|
| Source | Physical infrastructure: network at a stadium gate. |
| Stimulus | Fault (omission): the connection between the gate and FanZone is lost during entry, for up to 2 hours. |
| Artifact | Communication channel between FanZone and the Gate Validation System. |
| Environment | Normal operation: match day, peak entry period. |
| Response | Mask the fault: the gate keeps validating tickets against its locally synchronized ticket list, logs scan results locally, and synchronizes them when the connection returns. |
| Response measure | No downtime at the gate: each scan answered in under 1s during the outage.<br>0 valid tickets wrongly rejected.<br>Local scan results synchronized within 5 minutes after reconnection. |

### QAS-04: Forged or duplicated tickets

| Part | Security (integrity) |
|---|---|
| Source | Unknown user: a malicious fan or third party. |
| Stimulus | Attack to access a service: presents a forged QR code, an altered ticket, or a copy of a ticket that has already been scanned. |
| Artifact | Services and data: ticket issuing and gate validation. |
| Environment | Normal operation: online or offline (gate connected or disconnected) on match day. |
| Response | Detect and resist the attack: verify the ticket's digital signature and status, reject invalid or reused tickets, log the attempt with gate, time and ticket ID, and alert support/admin on repeated attempts. |
| Response measure | Probability of detection: 100% of forged, altered or reused tickets rejected in security tests.<br>While offline: 100% of tickets reused at the same gate rejected; tickets reused across different gates flagged within 5 minutes of reconnection.<br>Time: rejection shown to the gate operator in under 1 s.<br>Every rejection appears in the audit log. |

### QAS-05: Unauthorized access to personal data

| Part | Security (confidentiality) |
|---|---|
| Source | Known user: an authenticated Marketing Team member. |
| Stimulus | Attempt to display information: requests or exports personal data (name, email, phone, ID) of fans who have not been given marketing consent. |
| Artifact | Data: fans' personal data and the marketing functions. |
| Environment | Normal operation: Online |
| Response | Authorize and log: access to non-consented personal data is denied, only anonymized aggregate statistics are returned, and the attempt is recorded in the audit log. |
| Response measure | Prevention: 0 personal records of non-consenting fans returned in access-control tests.<br>Probability of detection: 100% of personal-data access attempts logged.<br>Time: the denial and the anonymized result are returned within 2s. |

### QAS-06: Friend authorization flow

| Part | Usability |
|---|---|
| Source | End user: a registered fan receiving an authorization request for the first time. |
| Stimulus | Wants to learn the system's features: opens the request email for the first time to approve or reject a friend's purchase. |
| Artifact | User interface: authorization page and email notification. |
| Environment | First use, normal operation, mobile or desktop browser. |
| Response | Provide feedback and assistance: the page shows who is buying, which match, ticket category and price, with clear Approve and Reject buttons; the buyer is notified of the decision. |
| Response measure | Task time: 90% of at least 10 test users complete the decision without help in under 60s from opening the email.<br>Steps: no more than 3 actions after opening the email if logged in.<br>Error rate: less than 5% of test users choose the wrong option or have to go back.<br>Support: 0 support tickets raised.<br>Feedback: The buyer sees the updated status within 5s. |

### QAS-07: Payment provider change

| Part | Modifiability |
|---|---|
| Source | Developer. |
| Stimulus | A wish to extend the system: add a second payment gateway or replace the current one. |
| Artifact | Code: the checkout / payment module. |
| Environment | Design time. |
| Response | The change is made, code-reviewed, tested and deployed without unintended side effects; booking, authorization and ticketing logic stay unchanged. |
| Response measure | Cost: completed within 3 person-days. Items affected: only the payment adapter module and its configuration. Risk: all existing checkout tests pass with no regressions. |

### QAS-08: EPFCA schedule update

| Part | Interoperability |
|---|---|
| Source | A system: the EPFCA data source. |
| Stimulus | A request to exchange information: EPFCA publishes a new or changed match schedule, stadium information or ticket prices. |
| Artifact | The systems that interoperate: FanZone's data import and the EPFCA data source. |
| Environment | Systems known prior to runtime; normal operation. |
| Response | The data is correctly accepted and applied to matches, or rejected and logged when invalid (e.g., unknown stadium or missing price), and the admin is notified of rejected records. |
| Response measure | Information exchanges correctly processed: 100% of valid records applied within 15 minutes of publication (assumed feed frequency). Correctly rejected: 100% of invalid records rejected and logged, with no partial updates to matches. |

# 6. Constraints

## 6.1 Technical constraints

| ID | Constraint |
|---|---|
| C-01 | FanZone is a web application usable from desktop and mobile browsers; no native apps. |
| C-02 | The system has at least two separate deployable parts (frontend and backend, plus the gate validation client). |
| C-03 | The prototype runs on free or student tiers. Production sizing is addressed in the architecture, not the prototype. |
| C-04 | Payments must go through an external payment gateway; the prototype must use the gateway's sandbox/test mode. |
| C-05 | EPFCA data format and frequency are not controlled by FanZone. (assumed) |
| C-06 | The Gate Validation System is external; FanZone can only provide data and signed tickets to it, not change its hardware. |

## 6.2 Business and regulatory constraints

| ID | Constraint |
|---|---|
| C-07 | Ticket sales must follow EPFCA's official match schedules, stadium information, pricing and timelines, and national sports regulations. Releases cannot open before official publication. |
| C-08 | Personal data must be processed according to applicable Egyptian data-protection law; marketing use requires user consent. (assumed) |
| C-09 | FanZone must not store full card data; card handling stays with the payment gateway (industry practice per PCI DSS [4]). |
| C-10 | All prices and payments are in Egyptian Pounds (EGP). (assumed) |
| C-11 | The delegated group purchasing rules (FR-05 to FR-08) must be enforced by the system, not only by the user interface. |

## 6.3 Organizational constraints

| ID | Constraint |
|---|---|
| C-12 | Team of 6 students; about 12 weeks across four milestones (M1 due October 23rd, 2026). (timeline assumed) |
| C-13 | Technology stack limited to tools the team knows or can learn within the timeline (stack to be fixed in M2). (assumed) |

# 7. Architecturally Significant Requirements (ASRs)

## 7.1 Utility tree

| Quality attribute | Refinement | Scenario / requirement | (Value, Impact) |
|---|---|---|---|
| Performance | Release-rush latency and capacity | QAS-01: 200,000 concurrent fans; 95% < 2 s; 0 accepted bookings lost. | (H, H) |
| Availability | External payment failure | QAS-02: fault detected in ≤ 30 s; no lost or orphaned payments | (H, H) |
| Availability | Offline gate operation | QAS-03: scans answered in < 500 ms during a 2-hour outage; 0 valid tickets rejected. | (H, M) |
| Security | Ticket authenticity | QAS-04: 100% of forged/reused tickets rejected in < 1s. | (H, H) |
| Security | Delegated purchase authorization | FR-05–FR-08, C-11: tickets for friends only with approved authorization. | (H, H) |
| Security | Confidentiality of personal data | QAS-05, C-08: 0 non-consented records returned; 100% of attempts logged. | (H, M) |
| Usability | Authorization flow | QAS-06: 90% decide in < 60s | (M, M) |
| Modifiability | Replaceable payment provider | QAS-07: new gateway in ≤ 3 person-days | (M, M) |
| Interoperability | Official data import | QAS-08, FR-13, C-05, C-07: EPFCA updates applied in ≤ 15 min | (M, M) |

## 7.2 Top ASRs and justification

| Rank | ASR | Derived from | Why it drives the architecture |
|---|---|---|---|
| 1 | **ASR-01: Zero overselling under release-rush load. Ticket counts per category must stay strictly consistent while 200,000 concurrent fans book with 95% of requests under 2 s.** | QAS-01, FR-04, FR-05, C-11 | Embodies the trade-off between performance/scalability and consistency. Shapes where inventory state lives, how concurrent writes are serialized (e.g., atomic reservations, queueing), what may be cached, and how the booking path scales. Getting it wrong means oversold stadiums: a safety and reputation failure. |
| 2 | **ASR-02: Server-side enforcement of delegated purchasing.** Authorizations follow Pending → Approved / Rejected, and checkout validates them atomically with payment confirmation. | FR-05–FR-08, C-11, QAS-06 | Core business differentiator. Requires a state machine with notifications and a consistency boundary across authorization, hold and payment; also drives the trade-off between security/privacy and usability. |
| 3 | **ASR-03: Reliable payment integration.** No charged-but-unbooked or booked-but-unpaid orders, even when the gateway fails or responds late. | QAS-02, QAS-07, FR-09, C-04, C-09 | Requires asynchronous callbacks, idempotent payment handling, timeouts, retries and reconciliation, and isolation of the gateway behind an adapter so it can be replaced (QAS-07). |

| Rank | ASR | Derived from | Why it drives the architecture |
|---|---|---|---|
| 4 | **ASR-04: Tamper-proof, offline-tolerant ticket validation.** 99.9% of forged or reused tickets rejected; gates keep validating offline in under 500 ms. | QAS-03, QAS-04, FR-10, FR-11, C-06 | Drives signed-ticket design, key management, ticket-list synchronization with the external gate system, and the trade-off between offline availability and duplicate detection. |
| 5 | **ASR-05: Consent-based privacy enforcement.** Marketing and support access personal data only according to role and consent; all access is audited. | QAS-05, C-08, FR-14, FR-15 | Affects data partitioning between operational and marketing data, role-based access control, anonymized analytics and audit logging. Lower impact than ASR-01–04 because marketing is out of the prototype, but high legal value. |